

Name
Class
R. No.

• Mathematics • +1 CBSE / IIT JEE
• Test 5(August 03, 2010) • Relations
• Maximum Marks 100 • Time 60 minutes

Instructions

1. The test contains nine subjective questions
2. The marking scheme will include marks for partially correct answers also.

Q1: Let R and R' be two relations in a set A . Prove the following:

- i) R and R' both are symmetric $\Rightarrow R \cup R', R \cap R'$ both are symmetric .
- ii) R and R' both are transitive $\Rightarrow R \cap R'$ is transitive but $R \cup R'$ needs not be transitive.
- iii) R and R' both are reflexive $\Rightarrow R \cup R', R \cap R'$ both are reflexive .
- iv) R is symmetric $\Rightarrow R^{-1}$ is symmetric and $R^{-1} = R$

20 Marks

Q2: Show that the relation R in the set of real numbers defined by $R = \{(a, b) : \sqrt{a} + 3 \leq b\}$ is neither reflexive, nor symmetric, nor transitive.

10 Marks

Q3: Let R be a relation in the set N . Let $R = \{(x, y) \in N \times N : x + 2y = 10\}$. Find the domain and range of R and R^{-1} .

4 Marks

Q4: A relation R is defined on the set $N \times N$, where N is the set of natural numbers, by setting $(a, b)R(c, d)$ if and only if $a + d = b + c$. Show that the relation is an equivalence relation. Describe the equivalence class of $(2, 3) \in N \times N$.

8+4 Marks

Q5: If R and R' be two equivalence relations on a set A . Prove that

- i) $R \cap R'$ is an equivalence relation in A .
- ii) $R \cup R'$ needs not be an equivalence relation in A .

14 Marks

Q6: Let $A = \{1, 2, 3, \dots, 14, 15\}$. Let R be the equivalence relation on A defined by congruence modulo 4.

- i) Find the equivalence classes determined by R .
- ii) Find a system B of equivalence class representatives which are multiples of 3.

3+2 Marks

Q7 : Each of the following expressions defines a relation on $\mathbb{R}$

- i) $x^2 - 4y^2 \geq 9$
- ii) $x^2 + 4y^2 \leq 16$

Sketch each relation in the plane $\mathbb{R}^2$.

10 Marks

Q8: Give an example of a relation , Which is

- i) Symmetric but neither reflexive nor transitive.
- ii) Transitive but neither reflexive nor symmetric.
- iii) Reflexive and symmetric but not transitive.
- iv) Reflexive and transitive but not symmetric.
- v) Symmetric and transitive but not reflexive.
- vi) Both symmetric and antisymmetric
- vii) Neither symmetric nor antisymmetric

14 Marks

Q9 Show that the relation R in the set A of points in a plane given by $R = \{(P, Q) : \text{distance of the point } P \text{ from the origin is same as the distance of the point } Q \text{ from the origin}\}$, is an equivalence relation. Further, show that the set of all points related to a point P (not the origin) is the circle passing through P with origin as the centre.

6+5 Marks